

Credit Card Fraud Detection

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Explore the research by following the link below:

<https://github.com/Niyal23/Fraud-Detection-Model>

Abstract

Financial fraud, especially credit card fraud, is a significant concern in the modern digital economy. This research project aims to develop a robust fraud detection system using machine learning techniques. Various models were trained and evaluated based on critical classification metrics to ensure high fraud detection rates while minimizing false positives. The dataset contains 1,000,000 transactions, and our analysis focuses on feature engineering, exploratory data analysis (EDA), model evaluation, and future improvements.

1. Introduction

Fraudulent transactions pose severe risks to financial institutions and consumers. Identifying fraudulent behavior requires analyzing complex transactional data patterns. Machine learning offers a viable solution to automate fraud detection, improving efficiency and accuracy compared to traditional rule-based systems.

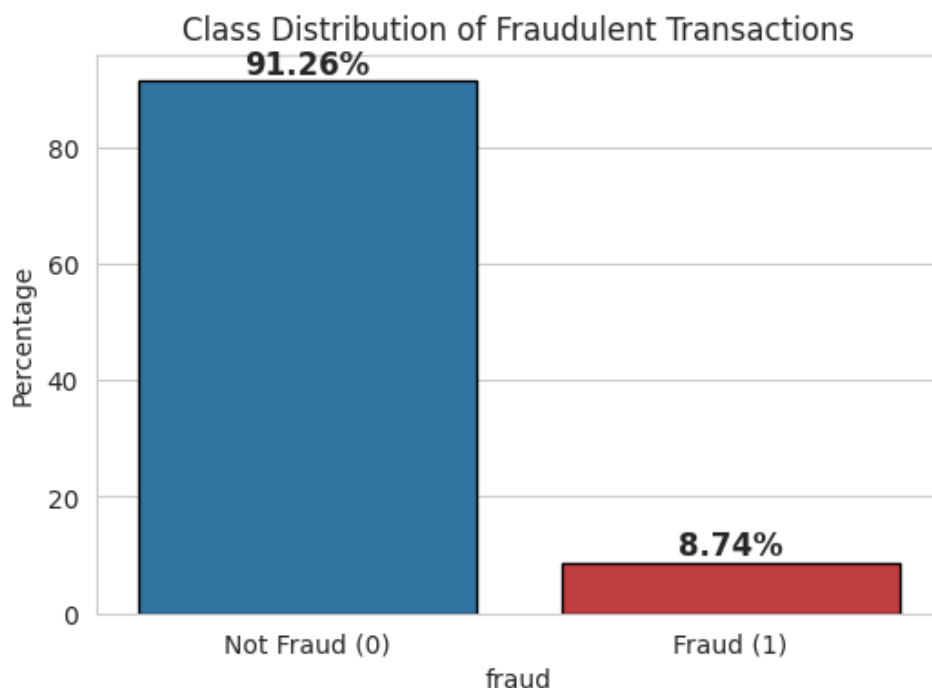
2. Data Exploration and Preprocessing

2.1 Dataset Overview

- The dataset contains **1,000,000 transactions** with 8 key features.
- No missing, duplicate, or infinite values were found.
- Class imbalance is evident, with fraudulent transactions comprising only **8.74%** of the total dataset.

Fraud Class Distribution

The dataset is highly imbalanced, necessitating techniques such as resampling or weighted modeling.



2.2 Feature Engineering

To improve predictive power, we applied several transformations:

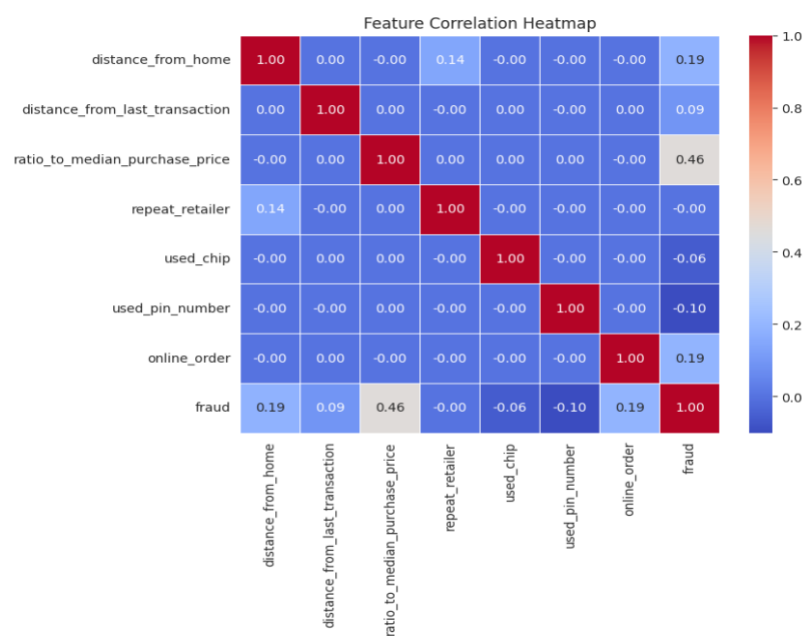
- **Log Transformation:** Applied to skewed continuous variables to normalize distributions.
- **Interaction Features:** Created new variables, such as distance_ratio and chip_pin_usage, to enhance model learning.
- **Feature Scaling:** Standardized numerical features for optimal model convergence.

2.3 Feature Correlation Analysis

A correlation matrix was computed to analyze relationships between features and fraud occurrence.

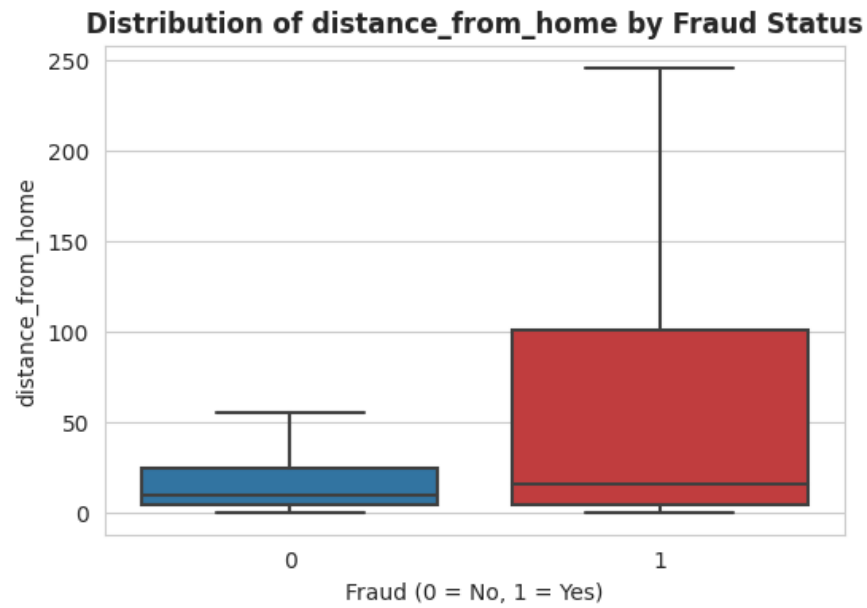
Key Observations:

- **ratio_to_median_purchase_price** shows the strongest correlation (0.46) with fraud.
- **distance_from_home** and **online_order** exhibit moderate positive correlations.
- **used_chip** and **used_pin_number** are negatively correlated with fraud, meaning transactions without these security features are more likely to be fraudulent.



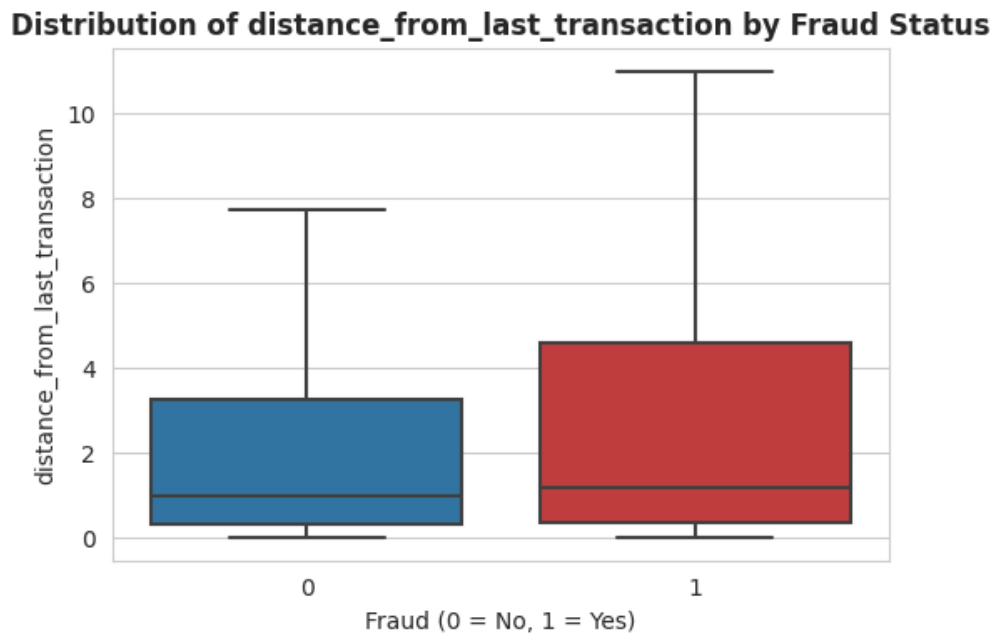
2.4 Feature Distributions by Fraud Status

Distance from Home



Fraudulent transactions tend to have a wider spread and higher values for **distance_from_home**.

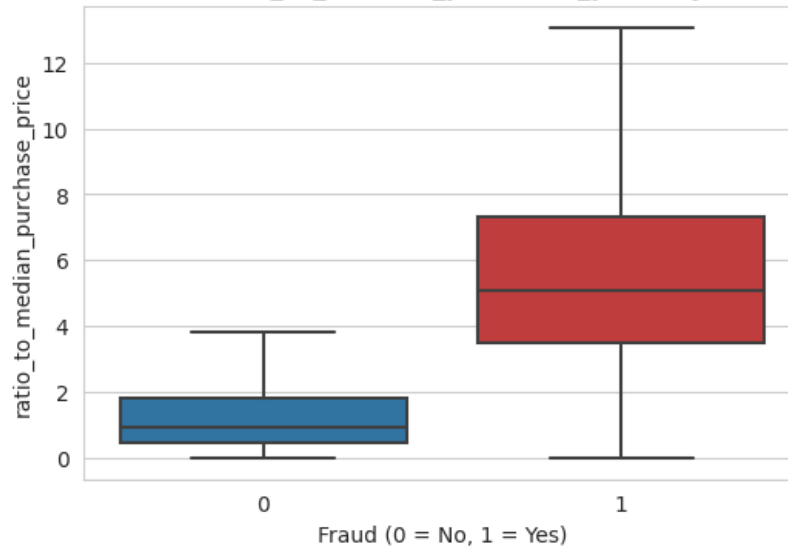
Distance from Last Transaction



Fraudulent transactions generally have higher **distance_from_last_transaction** values.

Ratio to Median Purchase Price

Distribution of ratio_to_median_purchase_price by Fraud Status



Fraudulent transactions exhibit significantly higher deviations from typical purchase amounts.

3. Model Development and Evaluation

We implemented and tested multiple machine learning models:

- **Logistic Regression**
- **Random Forest**
- **XGBoost**

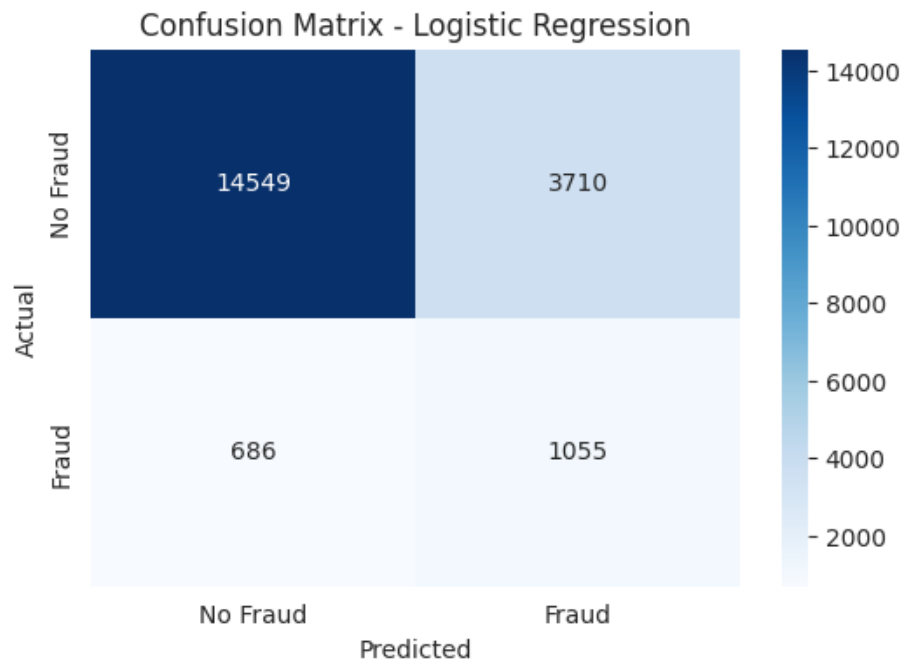
3.1 Model Performance Comparison

Model	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.221	0.606	0.324	0.818
Random Forest	1.000	0.999	0.999	1.000
XGBoost	0.993	0.993	0.993	0.999

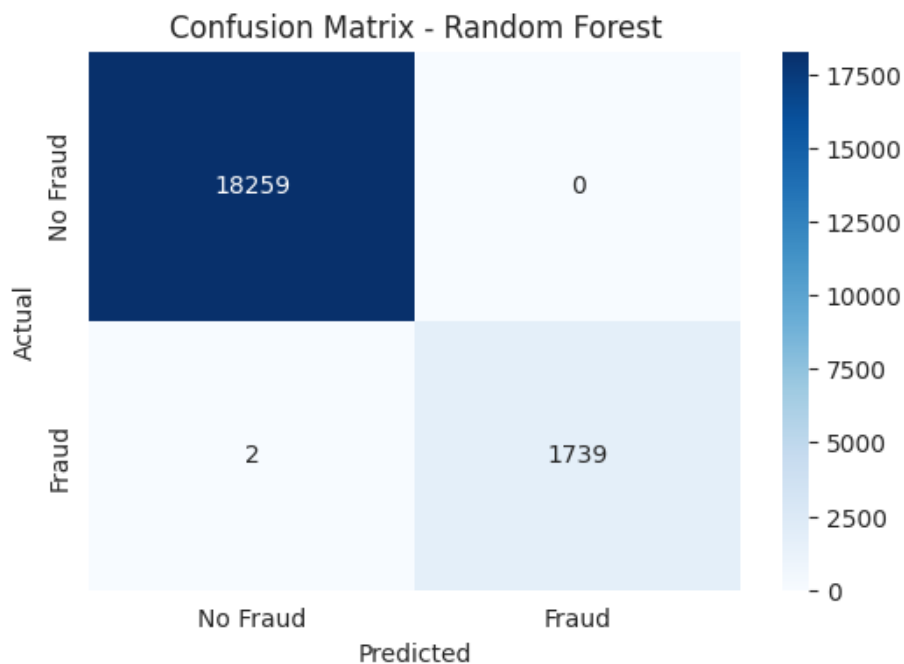
3.2 Confusion Matrices

To further evaluate model performance, we analyzed confusion matrices.

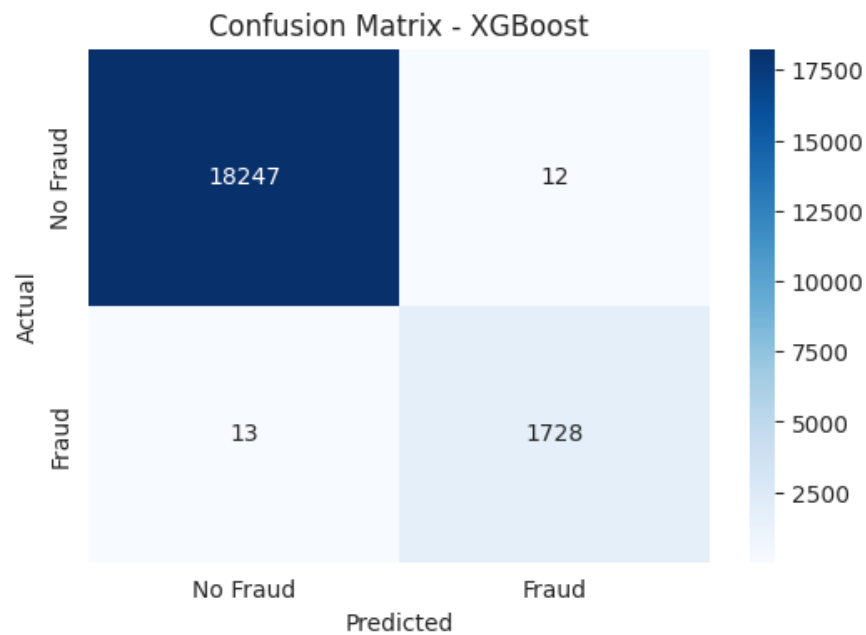
Logistic Regression



Random Forest



XGBoost



4. Conclusion

This research successfully implemented a fraud detection model using machine learning. Key findings include:

- The **Random Forest and XGBoost models** significantly outperform Logistic Regression, achieving near-perfect accuracy.
- The strongest predictor of fraud is **ratio_to_median_purchase_price**.
- Fraud detection improves when incorporating feature engineering techniques, such as log transformations and interaction features.